

Lab 19: Understanding & Implementing Sutherland-Hodgeman Polygon Clipping Algorithm

Assignment Date: February 12th, 2026

Deadline: February 21st, 2026

Class: CSIT 3rd Semester, New Summit College

Subject: Computer Graphics

Submission Mode: **Handwritten Document** (Print Output Section only)

Objective: Understanding & Implementing Sutherland-Hodgeman Polygon Clipping Algorithm

Tasks:

1. Understand the basics of the Sutherland-Hodgeman Polygon Clipping Algorithm
2. Write a Program for the Sutherland-Hodgeman Polygon Clipping Algorithm
3. Analyze the output and prepare a conclusion.

Report Format & Structure (Topics to include for submission):

- Index Page
- Theory:
 - a. Sutherland-Hodgeman Polygon Clipping Algorithm
 - b. Algorithm of Sutherland-Hodgeman Polygon Clipping Algorithm
- Program for Sutherland-Hodgeman Polygon Clipping Algorithm
- Implementation/Output (You need to label the **Before Clipping** and **Clipped Image with coordinates** clearly in the output image presented)
- Conclusion

References:

1. https://drive.google.com/file/d/1GyBfu8PCB4GBuOaHXPG4vueOYIVOm_Eu/view?usp=sharing
2. <https://www.geeksforgeeks.org/polygon-clipping-sutherland-hodgman-algorithm/>